

Question				Marking details		Marks available												
						AO1	AO2	AO3	Total	Maths	Prac							
	(b)	(i)		<table><tr><td>Part of the energy profile</td><td>Letter</td></tr><tr><td>energy change for the reaction</td><td>C</td></tr><tr><td>energy of the reactants</td><td>A</td></tr><tr><td>activation energy of the reaction</td><td>B</td></tr></table> award (2) for all three correct award (1) for any one correct	Part of the energy profile	Letter	energy change for the reaction	C	energy of the reactants	A	activation energy of the reaction	B	2			2		
Part of the energy profile	Letter																	
energy change for the reaction	C																	
energy of the reactants	A																	
activation energy of the reaction	B																	
		(ii)		the (minimum) energy required for a reaction to happen / start accept 'the <u>minimum</u> energy required to activate the reaction' neutral answer – the energy required to activate the reaction	1			1										
		(iii)		award (1) for any of following • the energy of the products is lower than the energy of the reactants • the product line is below the reactant line / E is below A • energy given out is greater than energy taken in / D is greater than B • lower energy at the end than at the beginning neutral answer – negative energy change		1		1										
				Question 7/1 total	4	3	2	9	0	3								